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A crossover-based multi-objective discrete particle swarm optimization model for solving multi-modal routing problems

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ABSTRACT

Passengers must use a set of modes and vehicles to reach their destination in complicated urban structures. Choosing an optimal route is a complicated optimization problem for these passengers. This study proposes a multi-objective optimization algorithm to solve the routing problem in the urban multi-modal network. The multi-modal network problem considered in this study is a transportation network with subway, Bus Rapid Transit (BRT), taxi, and walking modes. The objective functions determine the optimized route by considering route length, traffic, comfort, and safety. We develop a Crossover-Based Multi-Objective Discrete Particle Swarm Optimization (CBMODPSO) to solve the problem. CBMODPSO has been improved using mutation and crossover operators. Multi-Objective Artificial Bee Colony (MOABC), Multi-Objective Ant Colony Optimization (MOACO), Multi-Objective Biogeography-Based Optimization (MOBBO), Multi-Objective Gray Wolf Optimization (MOGWO), and Non-dominated Sorting Genetic Algorithm-II (NSGA-II) algorithms are used to evaluate and compare the results from the CBMODPSO algorithm. In addition, CBMODPSO results are compared with previous research results. We show the improved algorithm is more repeatable than other algorithms. The CBMODPSO algorithm has a faster convergence rate and is able to get to an optimal solution in a smaller generation number and much less time. The CBMODPSO algorithm is implemented in about one-thirties of the MOBBO duration. Meanwhile, it has a reproducibility of almost twice the MOGWO algorithm.

1. Introduction

Metaheuristic optimization has gained popularity among researchers in recent decades [1]. Studies in this field move more quickly because techniques are introduced mathematically and made available to everyone. These methods are aimed at increasing the effectiveness of existing solutions [2]. Single-objective and multi-objective optimization strategies are the two main metaheuristic optimization strategies. Genetic Algorithm (GA) [3], Stochastic Paint Optimizer (SPO) [4], Capuchin Search Algorithm (CSA) [5], Ant Colony Optimization (ACO) [6], Particle Swarm Optimizer (PSO) [7], Dynamic Water Strider Algorithm (DWSA) [8], Artificial Gorilla Troops Optimizer (AGTO) [9], Quadratic Interpolation Optimization (QIO) [10], Red Deer Algorithm (RDA) [11] Cuckoo Search and Stochastic Paint Optimizer (CSSPO) [12] are some of the recent single metaheuristic algorithms. The metaheuristic algorithms are applied to different fields of study [13–15]. In addition, Multi-objective Bonobo Optimizer (MOBO) [16], Advanced Neural Network Algorithm (ANNA) [17], Multi-objective material generation algorithm (MOMGA) [18], and Multi-Objective Crystal Structure Algorithm (MOCryStAl) [19] are some multi-objective metaheuristic algorithms.

Routing is among the classic issues, which was officially first dealt with by Hamilton titled the traveling salesman [20]. Nowadays, urban transportation is transported within a multi-modal transportation network in order to ease intra-city travel. This consequently has made routing difficult for urban travelers. Multi-modal transportation networks can encompass subway, bus, taxi, etc., modes [21]. The desired route might differ for every traveler in the urban multi-modal network. Travelers consider more than just the shortest route as a goal for route selection, as they consider other objectives such as traffic, comfort, and travel safety to reach their destination. More convenient routes might be less safe, which represents a misalignment of objectives. Due to

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different involving objectives and the misalignment of the objectives, multi-modal routing is a multi-objective optimization problem [22,23].

The multi-objective routing problem, even in the mono-objective case, is a complicated optimization problem of the Non-deterministic polynomial-hard (NP-Hard) order because it includes a wide variety of possible solutions [24]. In this problem, the search space is very wide due to numerous possible solutions; therefore, classical methods cannot solve this sort of problem in a reasonable period [25]. Many studies have been carried out on optimization algorithms due to the need to solve complicated problems with a wide search space [26]. Inspired by nature, meta-heuristic algorithms based on population and swarm intelligence have been developed to solve complex optimization problems [27]. The particle swarm optimization algorithm is among the population and swarms' intelligence-based algorithms, which has shown a strong performance in solving the problems [28,29]. This algorithm is dedicated to solving continuous problems; however, it exhibits excellent performance for discrete problems, including multi-modal routing problems, by applying some changes and generalizing them to the discrete condition [30].

Today, public transportation systems have made great progress in cities, which has made the public transportation system more complicated. Increased traffic, noise pollution, air pollution, reduced people's time, the need for speeding in transport, and congestion have augmented the need for public transportation systems. The existence of multi-modal transportation systems could improve service quality and solve urban public transportation problems [31]. Therefore, in these systems, passengers would need to use several transportation modes on the way to get to their destination. The studied investigations have mostly focused on mono-modal routing, and the combination of transportation modes has not been the case too much. The widely used particle swarm optimization algorithm is for continuous spaces. It has also been used for problems including the traveling salesman as well as mono-objective routing problems. The positive results from solving similar and simpler problems using particle swarm optimization algorithms indicate the increasing need for research activities to solve more complicated problems discretely and in the multi-objective form using a particle swarm optimization algorithm [32,33].

Much research has been done in order to solve the routing problem and find the optimal route [34,35]. The simplest case of the routing problem is when the network is in mono-modal and mono-objective conditions. The approaches used for routing problem solutions are exact and stochastic. In exact methods, one accurately correct solution is calculated at the end, but in stochastic methods, a set of solutions is presented for the intended problem, and close to optimal solutions (not completely optimal) will be generated. Among the popular definitive methods for solving routing problems, Dijkstra could be mentioned, on which lots of studies have been carried out [36].

Kheirikharzar et al. [37] solved a multi-modal routing problem using the Dijkstra algorithm by considering three objective functions, including cost, time, and the number of mode changes. They used the Pareto solution set and ran the algorithm once for each objective. The classic Dijkstra algorithm is not suitable for solving routing problems with an extensive transportation network in a reasonable time. Other accurate methods, such as A*, Depth-first Search (DFS), Breadth-first Search (BFS), Bellman–Ford, etc., are also available, where lots of research has been done. The basis of all these methods is the weighted graph network, and one route will be returned at the end [38]. Using classic algorithms for solving routing problems in different cases has led to dissatisfaction, especially when the routing problem contains a large network with bulky data [39].

Fuzzy logic has been used to solve multi-modal routing problems in lots of research [40,41]. In order to solve the problems using fuzzy logic, the accuracy of the gained solutions depends on the experience, fuzzy logic, and the modeling knowledge level of the person [26]. Furthermore, in large multi-modal networks, including a network as extensive as the whole of Tehran city, the solution to the problem would be more complicated due to the bulky multi-modal network [42]. Accordingly, considering the mentioned problems, part of the routing problem field studies has shifted to using meta-heuristic algorithms.

In order to solve the routing problem using the particle swarm optimization (PSO) algorithm, much research, such as [43] has been done. The routing problem is supposed to be mono-modal and mono-objective in these researches; in addition, the inherent of the routing problem is discrete, while the PSO algorithm is continuous. Eberhart discreetly presented the basic PSO algorithm using a binary statement [44]. The routing problem in the urban multi-modal network contains a complicated space; however, using binary coding for modeling this problem greatly adds to the complexity of calculations [45]. Thus, using the PSO binary is unsuitable for routing in large networks [46]. Pan et al. [47] promoted the PSO algorithm to discrete in a mono-objective manner. The movement and position update notions are conducted using the crossover and mutation to solve the no-wait flow shop scheduling problem. Zhang et al. [48] promoted the continuous particle swarm optimization algorithm to discrete in order to solve the traveling salesman problem. They named the new discrete particle swarm optimization integer and categorical PSO (ICPSO). Strasser et al. [49] presented the PSO edition in the conducted investigation, which can optimize the discrete variables. Recently, Zare et al. have published a new version of PSO, which enhanced the PSO algorithm based on Hummingbird Flight (HBF) patterns to improve the search quality and population diversity [50].

Optimization problems are divided into mono-objective and multi-objective classes. In many cases of multi-modal transportation, the passenger pursues several objectives, which may not be in the same direction and neither have a linear relationship with each other. The misaligned objectives include the shortest route with the lowest cost or the lowest-cost route with the least traffic volume. Most of the conducted investigations are concentrated on a single objective, which causes ignorance of other aspects of the routing [51]. Focusing merely on the shortest route leads to the ignorance of other route criteria, and perhaps the shortest route is the most time-consuming. The principle is to minimize more than one objective function, i.e., the problem should be solved based on a multi-objective decision-making [52].

One of the multi-objective algorithms' most popular and significant optimizations is the Non-dominated Sorting Genetic Algorithm II (NSGA-II) [53] algorithm. It is different from the genetic algorithm because it is a multi-objective one. NSGA-II is an elite evolutionary algorithm presented by Bob et al. [53]. NSGA-II is not limited in the number of objectives for optimization, and the crowding distance concept in the algorithm has made it superior compared to genetic multi-objective algorithms [54]. The main priority of the PSO algorithm, compared to other algorithms, including genetic and ant algorithms [55], is flexibility and its accessible operators, in general [55]. The multi-objective case of the PSO algorithm, multi-objective particle swarm optimization (MOPSO), was first developed by Coello et al. [56]. The output of the MOPSO algorithm Pareto of fronts of optimized solutions, and the user is able to choose the best solution according to his priorities. Several investigations and studies for the multi-objective use of the PSO algorithm will be discussed.

In 2012, Masoumi et al. [57] solved the land-use allocation problem using the MOPSO algorithm. Using mapping, they solved the discrete problem using the continuous MOPSO algorithm. In this study, there are 34 land-use, which are named from 1 to 34. Using the bracket plus 1 (ceiling) function, Masoumi et al. [57] rendered continuous values from 1 to 34. Discretization of continuous values using this function in the MOPSO algorithm returns incorrect and false values. For example, a particle of value 3.8 would be rounded to 4 (indicating the number of stations) using the rounding function, and the position of station 4 might be far away from the coordinate position of 3.8. accordingly, applying the mapping operation between the discrete space of the problem and the continuous search space of the MOPSO algorithm, using the rounding function used in this work, cannot be used to solve discrete multi-modal routing problems [57].

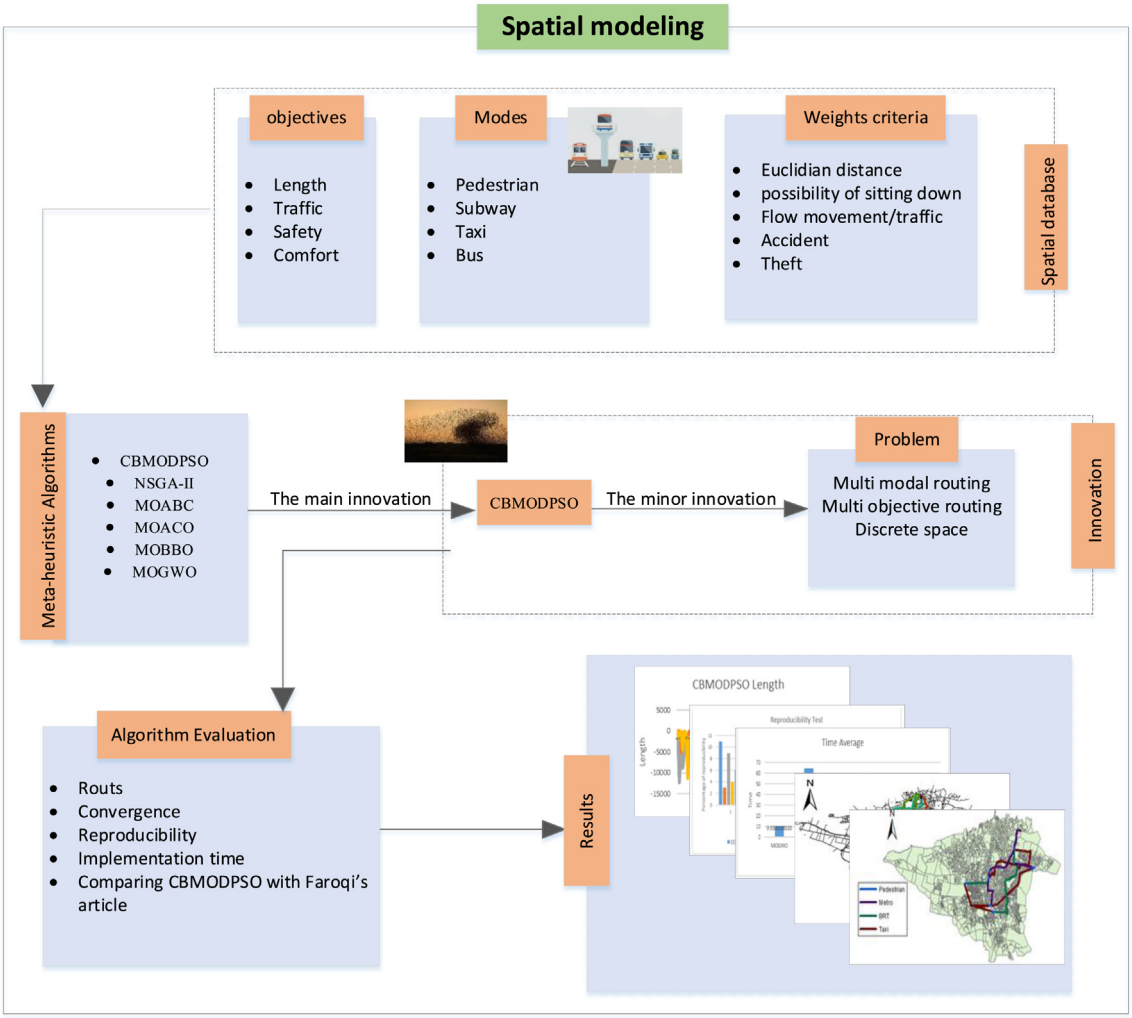

Fig. 1. Steps of research.

In the present study, the multi-objective routing problem is in an urban multi-modal transportation network as extensive as the Tehran metropolis multi-modal transportation network. Subway, BRT, walking, and taxi modes and four objective functions of traffic, distance, safety, and comfort have been assumed. Besides, the routing problem has been solved in a multi-objective way, and Pareto the front has been used. In order to solve this discrete problem, particles' discrete search motion was used in the multi-objective particle swarm algorithm. So, this research has promoted the MOPSO algorithm to the discrete condition. The results from the promoted algorithm are evaluated and compared with Multi-Objective Ant Colony Optimization (MOACO) [58], Multi-Objective Biogeography-Based Optimization (MOBBO) [59], Multi-Objective Artificial Bee Colony (MOABC) [60], Multi-Objective Gray Wolf Optimization (MOGWO) [61], and NSGA-II [62] algorithms as well as the multi-pheromone ant colony optimization (MPACO) and multi-colony ant colony optimization (MCACO) [24]. The main reason to choose these algorithms is that the algorithms are discrete inherently, so, these are suitable to solve the problem. Besides, different research shows that these are suitable for routing problems.

Section 2 shows the multi-modal routing problem and Crossover Based Multi-Objective Discrete Particle Swarm Optimization (CBMODPSO). In Section 3, the problem modeling method and the used data will be presented. The results from the developed algorithm will be compared with MOACO, MOBBO, MOABC, MOGWO, and NSGA-II algorithms in Section 4. Finally, the conclusions of this study are presented in Section 5.

2. Materials and methods

2.1. Methodology

The present study to conduct spatial modeling of multi-modal routing in the urban environment included four steps shown in Fig. 1. In the first step, the database includes objectives, modes, and weights criteria. In the second step, the innovation has been developed, including primary and minor innovations the main innovation related to enhancing MOPSO, and the minor innovation includes a multi-modal routing problem. In the third step, the routing problem was implemented and showed results. Eventually, the algorithm (CBMODPSO) evaluation was conducted by comparing NSGA-II, MOABC, MOACO, MOBBO, and MOGWO.

2.2. Multi-modal routing problem

These days, human transport in urban areas is often through a multi-modal network. People can use multiple transportation modes during their travel [42]. However, the transportation system has become more complicated now. Thereby, in order to help people find optimal or close to optimal routes through complicated transportation network development, route planning in the multi-modal network has gained considerable importance [63]. Multi-modal transportation makes better use of ground, air, and energy resources in comparison with mono-modal transportation [42].

$$\begin{bmatrix} 0 & 1 & 1 & 0 & 1 & 1 \\ 1 & 0 & 1 & 1 & 1 & 1 \\ 1 & 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 0 & 1 & 0 \\ 1 & 1 & 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 & 1 & 0 \end{bmatrix} \quad \begin{bmatrix} 0 & 1 & 2 & 0 & 1 & 3 \\ 1 & 0 & 1 & 4 & 3 & 1 \\ 2 & 1 & 0 & 0 & 4 & 1 \\ 0 & 4 & 0 & 0 & 2 & 0 \\ 1 & 3 & 4 & 2 & 0 & 1 \\ 3 & 1 & 1 & 0 & 1 & 0 \end{bmatrix}$$

a) b)

Fig. 2. Multi-modal matrix. (a) Adjacent matrix, number 1 indicates a relationship between two nodes (stations) and number 0 indicates a lack of relationship between two nodes (stations); (b) Mode type matrix, each of the numbers 1, 2, 3, and 4 indicate the sort of the corresponding edge mode.

A multi-modal route consists of a set of mono-modal routes. Juxtaposition and sequence of mono-modal routes, as well as taking constraints and problem limitations into account, the multi-modal routing is achieved [64]. In general, modeling transportation systems is challenging work, especially when the number of modes increases. Multi-modal transportation system is a combination of all traveling cases and different types of transportation systems working through other systems, namely, a collection of transportation methods that passengers can use to reach their destination [42].

Graphs are typically used to model a multi-modal network. In a multi-modal graph, an adjacent matrix is used to determine whether there is an edge between two stations (nodes). Meanwhile, the mode of each edge is introduced using a label matrix. Each edge holds its weight [65]. An example of a multi-modal network graph is shown in Fig. 2. In Fig. 2b; the label matrix includes four modes of BRT, subway, walking, and taxi. The graph adjacent matrix is also illustrated in Fig. 2a. The graph of the adjacent matrix and the label matrix of Fig. 2 is presented in Fig. 3.

2.3. CBMODPSO

The PSO method was first introduced by Kennedy and Eberhart [7] in order to solve mono-objective problems with continuous variables. In the particle swarm optimization algorithm, a space is assumed to be of n dimensions, each of which indicates some values which are assigned to a part of the problem solution. In other words, each dimension is equivalent to a gene in the genetic algorithm; namely, there would be one dimension (or one particle) in the n-dimension space of the PSO algorithm for each gene constituting chromosomes in the genetic algorithm. So, the complete solution to the problem is one point in the n-dimension space of the PSO algorithm. Chains of part solution values in the PSO algorithm occur as a position change or move in the n-dimension space of the problem. So, changes in initial random solutions towards optimal solutions occur as particle movement (or velocity) in the n-dimension space of the algorithm. PSO is interpreted as the movement of poorly positioned particles towards particles with optimal positions. In other words, every moving particle in the n-dimension space is equivalent to a changing solution and evolving towards better solutions. Every particle stores the best position it has had previously in terms of the objective function in the storage and becomes informed about the best experience of other particles [44].

In the PSO algorithm, particles have movement in the search space. The displacement of each of these particles is influenced by both factors of the particle's own experience and the position of other particles. Therefore, the position of each particle affects other particles. The modeling result of this social behavior is a searching process in which particles move toward optimal areas. The principle of the PSO algorithm is based on the fact that each particle sets its position at every moment in the search space according to the best position it has already been positioned into and the best position among all of the neighbors. Just like other evolutionary algorithms, the particle swarm algorithm starts with a random population development which is read as a group of particles here. The movement vector of every particle to achieve the best solution include the present position of the particle, the present velocity of the particle, a random movement vector, a vector between the current position and optimal personal position, and a vector between the current position and the optimal global position. Accordingly, the velocity of each particle and, in turn, its new position is according to Eq. (1).

$$\begin{aligned} V_i(t+1) = {} & !\, V_i(t) + c_1 \,\mathrm{rand}(0{:}\ 1)\ \ \mathrm{pbest}_i - X_i(t)\ \ + c_2 \,\mathrm{rand}(0{:}\ 1) \\ & \times\ \ \mathrm{gbest}(t) - X_i(t) \end{aligned} \tag{1}$$

$$X_i(t+1) = X_i(t) + V_i(t+1)$$

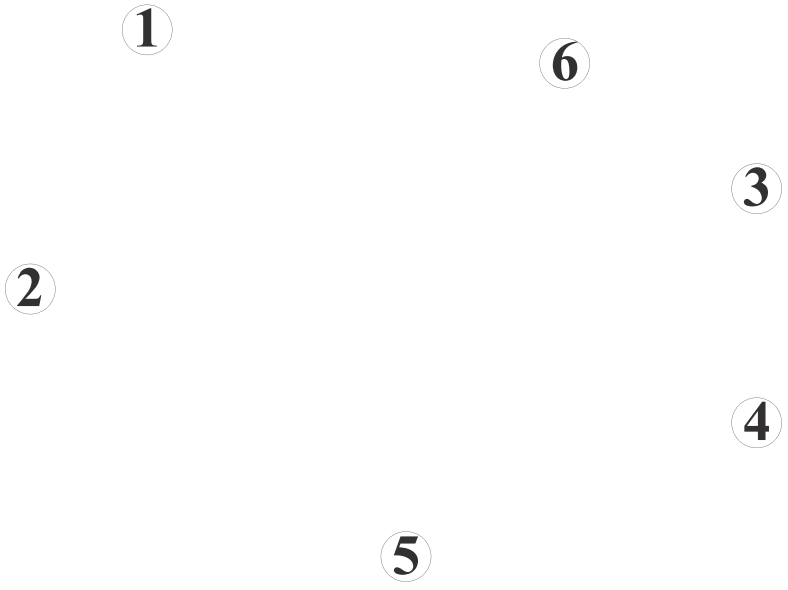

Fig. 3. An example of the multi-modal graph relating to Fig. 2 matrices. Edges with different colors indicate the different modes corresponding to each edge. The numbers inside the circles show the ID of each station. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)